ENALAPRIL

Teratogenic
Angioedema

CHEMICAL NAME (2S)-1-[(2S)-2-{[(2S)-1-ethoxy-1-oxo-4-phenylbutan-2-yl]amino}propanoyl]pyrrolidine-2-carboxylic acid (maleate)
CLASS ACE inhibitor (prodrug)
TRADE NAMES Renitec · Vasotec · Enap · Vasoretic (with HCTZ)

THERAPEUTIC USES / INDICATIONS

- Hypertension.
- Heart failure with reduced ejection fraction (↓ mortality).
- Asymptomatic LV dysfunction.
- Diabetic and chronic kidney disease with proteinuria.

ADMINISTRATION

- **PO** once or twice daily, with or without food.
- **IV** enalaprilat (where available) over 5 min.
- First dose at bedtime if on diuretics.

DOSAGES

- **Hypertension:** start 5 mg daily (2.5 mg if on diuretic/elderly/renal); usual 10–20 mg; max 40 mg/day.
- **Heart failure:** 2.5 mg twice daily, titrate to 10–20 mg twice daily.
- **Children:** 0.08 mg/kg once daily (max 5 mg), titrated.
- CrCl <30: start 2.5 mg.

PHARMACOKINETICS

- **Onset:** 1 h. **Peak:** 4–6 h. **Duration:** 12–24 h.
- Prodrug hydrolysed in liver to active enalaprilat.
- Excreted in urine. **t½:** 11 h (enalaprilat).

PHARMACODYNAMICS

- Inhibits **ACE** → ↓ angiotensin II and aldosterone → vasodilation, natriuresis.
- Reduces cardiac remodelling; ↓ intraglomerular pressure (renoprotective).

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Dry cough.
- Dizziness, headache, fatigue.
- Hypotension (first dose).

ADVERSE EFFECTS (SERIOUS)

- Angioedema.
- Hyperkalaemia.
- Acute kidney injury.
- Neutropenia (rare); hepatitis (rare).

CONTRAINDICATIONS & CAUTIONS

- Pregnancy.
- History of ACE inhibitor angioedema; hereditary angioedema.
- Bilateral renal artery stenosis.
- Within 36 h of sacubitril/valsartan.

KEY INTERACTIONS

- **Potassium supplements, spironolactone, trimethoprim:** hyperkalaemia.
- **NSAIDs:** ↓ effect, AKI.
- **Lithium:** toxicity.
- **Insulin, sulfonylureas:** ↑ hypoglycaemia.

NURSING MANAGEMENT

- Monitor BP, HR, weight and oedema (HF).
- Check U&E at baseline, 1–2 weeks, then periodically.
- Accept creatinine rise up to 30%; report more.
- Watch for angioedema.
- Hold if hypotensive or K^+ >5.5.

PATIENT EDUCATION

- Rise slowly from sitting.
- Report face/lip swelling, fainting, persistent cough.
- Avoid NSAIDs and potassium salt substitutes.
- Do not use in pregnancy.

KEY POINTS

- Prodrug ACEi; reduces mortality in HFrEF.
- Monitor K^+ and creatinine.
- Titrate to target dose.
- Cough → switch to ARB.